Formal companion to the Erdős 81 transfer note

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Formal assembly of the IV–V application
Preprint 1.0, October 18, 2026

This companion describes the final Lean source package for Series research note I, Preprint 1.0. The note reorganizes part of the proof in Paper IV through Paper V's transfer interface. It claims no new solution of Erdős Problem 81 and no mathematically independent proof. The statements and proof terms retain their established scope. The current validation is bound to the source manifest and final logs identified below.

1. What the formalization proves

The public conclusions produce actual clique partitions of the original graph. For all sufficiently large chordal graphs of order n, the partition has at most

$$M(n) = \left\lfloor \frac{n(n+1)}{6} \right\rfloor$$

blocks, each with at most four vertices. For all orders, including zero, an absolute additive constant suffices. There is no assertion of the sharp bound at every order, no numerical upper bound on the eventual threshold, and no new extremal value claimed by this application.

The source root is Erdos81Note.lean. It imports Erdos81Note.Assembly, which imports the physical transfer adapter and the required results of IV directly. The assembly invokes the actual realizable-class transfer theorem from V. It does not invoke IV's final all-chordal bound. The project remains mathematically dependent on the structural and rounding results of IV and on the transfer theorem of V; the separate project records a different assembly of those proved inputs. Paper VI is not an input.

The principal note keeps one paragraph on formalization. This companion gives the model boundary, the six checked public interfaces, the selected dependency chain, the complete local-project axiom audit, and reproduction information.

2. Graphs, resources and integer arithmetic

The physical graph model is `SimpleGraph V` with a finite vertex type, decidable equality and decidable adjacency. The first two public interfaces work on an arbitrary finite type. The final graph theorems use `Fin n`. Chordality is the predicate `PaperIV.IsChordal G`, which is printed as G.IsChordal in the endpoint gate.

A packing item is a finite set of vertices that is a clique and has cardinality three or four. Its resources are its unordered pairs of distinct vertices, hence actual graph edges. A packing consists of such items with disjoint edge supports. Its natural-valued gain is

$$g(P) = \sum_{Q \in P} \left(\binom{|Q|}{2} - 1 \right).$$

The adapter physicalCS makes two distinct items conflict precisely when their edge supports overlap. The maps toPacking and liftPacking prove the correspondence between feasible outputs of V and physical packings in IV; their gain lemmas preserve the weighted sum. This step rules out a certificate whose alleged realization is merely an auxiliary object without graph edges.

A `CliquePartition G` contains a finite family of cliques, each of order at least two, with pairwise disjoint edge supports whose union is exactly G.edgeFinset. Q.size counts blocks, and `Q.OrderAtMost 4` bounds their orders. Completion adds one two-vertex block for each uncovered edge and proves

$$|Q| + g(P) = e(G).$$

Thus the subtraction used in the paid-packing interface is natural subtraction with a proved bound $g(P) \le e(G)$; it is not an unexamined truncation. Likewise, `targetSize n` is natural division $n(n+1)/6$, exactly $M(n)$.

The rational predicate `CertifiedFractionalOptimum G w` means that there exist a feasible fractional mixed packing and a feasible dual cover, both of value w. It supplies matching primal and dual witnesses, rather than an unsupported assertion that a maximum exists. The assembly obtains this certificate from IV's proved existence theorem before splitting into the far and near cases.

3. The six checked public interfaces

All names in this section belong to the namespace Erdos81Note. The endpoint gate checks their full elaborated types and prints their axiom dependencies. The final output is retained as evidence/ENDPOINTS.log; the following descriptions retain every substantive hypothesis and conclusion.

3.1. partition_of_certificate

Let G be any finite simple graph, C an auxiliary conflict system, and let the weight function w, multiplicity function X, class count q, and target T be natural-valued. Assume $1 \le q$ and a certificate

$$\texttt{RealCert}\big(C, \mathrm{physicalCS}(G), w, \mathrm{physicalWeight}(G), X, q\big).$$

The certificate includes deleted occurrences, an assignment of the remaining occurrences to classes, and a feasible physical realizer for each class with an explicitly bounded natural-valued loss. Its cost is the deleted weight plus the sum of realization losses. If

$$q\, e(G) + \mathrm{cost}(\mathrm{cert}) \le qT + \mathrm{wMultSize}(C, w, X),$$

then there exists a `CliquePartition G` of order at most four and size at most T. No chordality hypothesis is needed. The proof calls PaperV.realizable_class_transfer, converts the selected realization into a physical packing, and completes it.

3.2. partition_of_paid_packing

For any finite graph G, natural target T, and physical mixed packing P, assume

$$e(G) - g(P) \leq T.$$

Then there exists a clique partition of G, of order at most four and size at most T. Internally, the proof chooses one auxiliary occurrence with reference weight $e(G)$, one class, and realization loss $e(G) - g(P)$, and applies the preceding interface. The construction records a paid physical packing in the transfer language; it does not claim an additional saving from averaging.

3.3. far_certificate

For every rational $\eta > 0$, there exists a natural threshold N such that, for every $n \geq N$, every graph G on `Fin n`, and every rational certified optimum w, the inequality

$$e(G) - w < \frac{n^2}{6} - \eta n^2$$

implies a clique partition of order at most four and size at most $M(n)$. This interface does **not** require chordality. It uses IV's uniform transfer with parameter $\eta/2$; the far-regime margin pays the rounding loss. Its threshold is existential, not the explicit near-branch threshold below.

3.4. near_certificate

Let $n \geq 4 \cdot 10^{12}$, let G on `Fin n` be chordal, and let $w \in \mathbb{Q}$ have a certified fractional optimum witness. Assume the negation of the strict far inequality:

$$\neg \left(e(G) - w < \frac{n^2}{6} - \eta n^2 \right), \qquad \eta = 10^{-16}.$$

Then there exists a clique partition of order at most four and size at most $M(n)$. The proof obtains an actual near-structure witness through IV's descent localization, uses its physical packing and retained accounts, proves that its completion is paid, and invokes partition_of_paid_packing. Neither a target partition bound nor a pre-existing root witness is assumed in this interface.

3.5. erdos81_eventual

There exists $N \in \mathbb{N}$ such that, for every $n \geq N$ and every chordal graph G on `Fin n`, there exists a clique partition of order at most four and size at most $M(n)$. The certified optimum and the case split are obtained internally. The chosen threshold is the maximum of the far-branch threshold at $\eta = 10^{-16}$ and $4 \cdot 10^{12}$; the latter alone is not asserted to suffice for the global theorem.

3.6. erdos81_all_orders

There exists $b \in \mathbb{N}$ such that, for every $n \in \mathbb{N}$ and every chordal graph G on `Fin n`, there exists a clique partition of order at most four and size at most $M(n) + b$. The proof takes $b = N^2$ from the previous theorem. Below the threshold it uses individual edges; above it, the eventual theorem. No positive-order assumption excludes the empty graph.

4. Which results are used, and which are excluded

The far branch calls PaperIV.RC01FarAssembly.uniformTransferAt. The near branch calls PaperIV.NearH1Localization.exists_nearStructureWitness_of_nearRegime and reaches the descent-localization declaration PaperIV.LocalStability.localize_original_of_windowed_accounts_descent. Optimum existence is supplied by PaperIV.CertifiedOptimumExistence.exists_certifiedFractionalOptimum.

The selected proof-chain audit begins with the certificate interface and the two final bounds. It follows declaration types and proof values through the local series and records 6,528 visited declarations. It requires actual dependencies on V's transfer and IV's descent localization. It rejects declarations originating in PaperIV.Erdos81Unconditional, the final global-bound module. The recorded forbidden list and normalized first-entry hits are empty.

This is a dependency claim about the selected proof, not a claim that all imported source files omit alternative arguments. Some upstream files contain unused first-entry developments. Their presence in a source package does not make those declarations premises of the final application. The complete dependency listing is preserved as DECLARATION_DEPENDENCIES.txt.

The physical partition is constructed once in the original graph after localization. The formalization does not transport partitions backwards along vertex-copy steps and does not charge an extra loss per step. The certificate theorem organizes the gain budget; the structural localization and physical construction remain the work of IV.

5. Final validation and trusted boundary

The canonical project uses leanprover/lean4:v4.35.0 and Mathlib commit 56a005855ea17e0b0958f73ec801ae140b0a1679. The source closure contains 324 local modules. The final validation consists of a root build, the six-interface gate, and a full local-project audit. The build uses complete, trace-checked cached artifacts for unchanged dependencies and recompiles the changed modules; it is not described as a fresh build of all upstream sources or of Mathlib.

The selected root audit traverses 6,528 declaration dependencies. It checks actual use of V's transfer and IV's descent localization and excludes IV's final global-bound module. The full audit covers 6,850 local declarations and visits 10,304 dependencies. These are distinct scopes: the latter includes the private helpers and other declarations in all imported local modules.

The full audit first verifies that every module in the source manifest is present in the imported Lean environment; a missing module causes rejection.

The permitted axiom union is propext, Classical.choice and Quot.sound. The gates reject additional mathematical axioms and sorryAx. The successful command exits and logs are recorded in evidence/BUILD_CHECKS.json; FINAL_FORMAL_VALIDATION.json binds those records to the exact source closure and the checks of the public entry point.

The author of the note's formalization is Juan Pablo Traverso Gianini. Its own modules and gates identify the author in their headers. AUTHORSHIP_NOTICE.md credits the inherited series source cuts without altering their existing mathematical and assistance attributions. The Paper V dependencies use the author-credited source version recorded in the manifest. The note's own modules and audit gates are distributed under Apache License 2.0, with the complete license in lean435/LICENSE and source attribution in lean435/NOTICE. Inherited license and authorship notices remain applicable.

lean435/OWNERSHIP_AND_LICENSE.json enumerates, with hashes, all author-owned Lean sources, including the reused series cuts, released by the author under Apache 2.0. The new

full headers cover this note's specific modules and audit entry points; inherited headers and mathematical attributions are preserved. External dependencies retain their own licenses.

6. Reproduction and immutable package

Extract NOTE_I_LEAN435_FINAL_SOURCE_v1.0.zip, enter lean435, and run:

```
lake exe cache get
lake build Erdos81Note
lake env lean -j 1 -s 32768 Endpoint435Gate.lean
lake env lean -j 1 -s 32768 FreezeFullAudit.lean
```

Preserve the supplied lake-manifest.json; the cache command retrieves artifacts for the pinned dependencies without requesting a new dependency selection.

The root build runs the selected dependency audit in Erdos81Note.lean. The endpoint gate prints the six public types and their axiom dependencies. The full audit checks every declaration in the imported local source closure. The archive includes all required local source files and pinned configuration; it excludes .lake caches and development junctions. The package does not require a checkout of Papers IV or V.

The source archive SHA-256 is:

`eae69a0316113f1701246b54243a1b51696b85ea656309319b8738932f90f604`

SOURCE_MANIFEST.json gives the hash of every packaged source and configuration file. Its own SHA-256 is:

`5f7deb401e26fcce100d576e4ed5c6975a46d14265144a1f267439b67f24613f`

FINAL_FORMAL_VALIDATION.json records the build, export and axiom results for this package. Exact-arithmetic regression scripts are supplementary checks, not premises of the universal theorems.

Assistance and scope

The principal note retains Paper IV's attribution of Claude, by Anthropic, ChatGPT/Codex, by OpenAI, Aristotle, by Harmonic, Certo and Jacobian for the inherited work. Aristotle's contribution to that work included exploration, counterexample search, proof development, formalization and review, and was not limited to translating finished proofs. For this note, ChatGPT/Codex assisted with the adapters, verification and presentation of the final package. Computations were reviewed within their stated scope; a finite search does not replace a universal proof. Kernel verification does not establish historical priority or replace external review.

The author remains responsible for the arguments, citations, code, and presentation. No AI system is listed as an author.